

EKLAKH DEWAN

AI Systems Engineer — Backend Systems — AI Infrastructure — Distributed Workflows
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SUMMARY

AI Systems Engineer focused on reliable backend infrastructure for AI-driven applications. Experienced in API architecture, asynchronous task processing, retrieval pipelines, persistence, guardrails, observability, CI/CD, and LLM workflow orchestration using Python, FastAPI, PostgreSQL, Redis, and Docker.

EDUCATION

KPRIET

B.Tech. in Artificial Intelligence & Data Science

2027

CGPA: 8.55

EXPERIENCE

Artificial Intelligence Intern

Flowrage Technology

Virtual Internship — 4 Weeks

- Integrated third-party AI APIs into application workflows and connected LLM capabilities with backend processing pipelines.
- Developed resume-processing and job-matching components involving text ingestion, transformation, similarity computation, and API-driven recommendations.
- Worked across AI service integration, data processing, API workflows, error handling, and application-level automation.

SELECTED SYSTEMS PROJECTS

APX — Accounts Payable Exception Resolution Agent

[GitHub](#)

Python, FastAPI, BM25, Dense Retrieval, Reranking, LLMs

- Architected an evidence-driven exception-resolution pipeline spanning deterministic validation, retrieval, agent, decision, and persistence stages.
- Combined BM25 and dense retrieval with hybrid fusion, cross-encoder reranking, and evidence-validity checks before downstream resolution decisions.
- Designed deterministic guardrails, persistence boundaries, and staged execution to keep AI-assisted workflows controlled and auditable.

TaxTrace — AI-Assisted Tax Reconciliation Platform

[GitHub](#)

FastAPI, Next.js, PostgreSQL, SQLAlchemy, Alembic

- Built a backend-driven tax reconciliation platform with structured exception review and notice-response workflows.
- Implemented relational persistence using PostgreSQL and SQLAlchemy with Alembic migrations for task, exception, notice-case, and draft entities.
- Developed a Next.js review workspace backed by FastAPI and React Query; Stage 6 backend validation reached **91 passing tests**.

AI-Powered Business Operations SaaS

[GitHub](#)

React, TypeScript, NestJS, PostgreSQL, Redis, BullMQ, Docker

- Designed a multi-tenant business operations architecture separating synchronous API requests from asynchronous AI workloads.
- Integrated Redis and BullMQ for background task processing and structured backend workflows around API, persistence, and AI-service boundaries.
- Containerized application services with Docker and designed the system around role-based access control and scalable service boundaries.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL

Backend: FastAPI, NestJS, REST APIs, PostgreSQL, SQLAlchemy, Node.js

Distributed Systems: Redis, BullMQ, Asynchronous Processing, Task Queues, Idempotency, Retry Handling

AI Infrastructure: RAG, LLM APIs, Agentic Workflows, Hybrid Retrieval, BM25, FAISS, Reranking

Infrastructure: Docker, GitHub Actions, CI/CD, Cloudflare Workers, API Integration

Frontend: React, Next.js, TypeScript, Vite

ENGINEERING FOCUS

System Architecture — API Design — Async Processing — Persistence — Guardrails — Observability — CI/CD — AI Workflow Orchestration